

Assignment 01 Computer Network

Name : Saurav Kumar

Roll No: D-25577

Registration Number : 25105142901

Semester : 4th Sem

Branch : CSE

Session : 2025 – 2028

College : Government Engineering College Kishanganj

Submission Date : 27-05-2026

Teacher Name: Prof. Lalan Kumar Sir

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point, and end devices (PCs, Laptops, etc.)

* Identifying the LAN Topology

On the basis of physical layout and structural inspection of the campus network at our organization, the LAN uses a Hierarchical Star Topology (also known as Extended Star Topology).

In this setup, the network is divided into three different logical layers:

Core Layer (Central Hub): The main server room is located on the 2nd Floor (Room 210). This room contains the main gateway/core router, firewalls, and central distribution switches that manages internet bandwidth and network security.

Distribution/Access Layer (Floor Hubs):

Each floor of the building is equipped with wall-mounted D-Link Network Racks (Cabinets). Inside these racks, TP-Link L2+ Managed Switches (Models: SG3428XMP & TL-SG1218MPE) are installed.

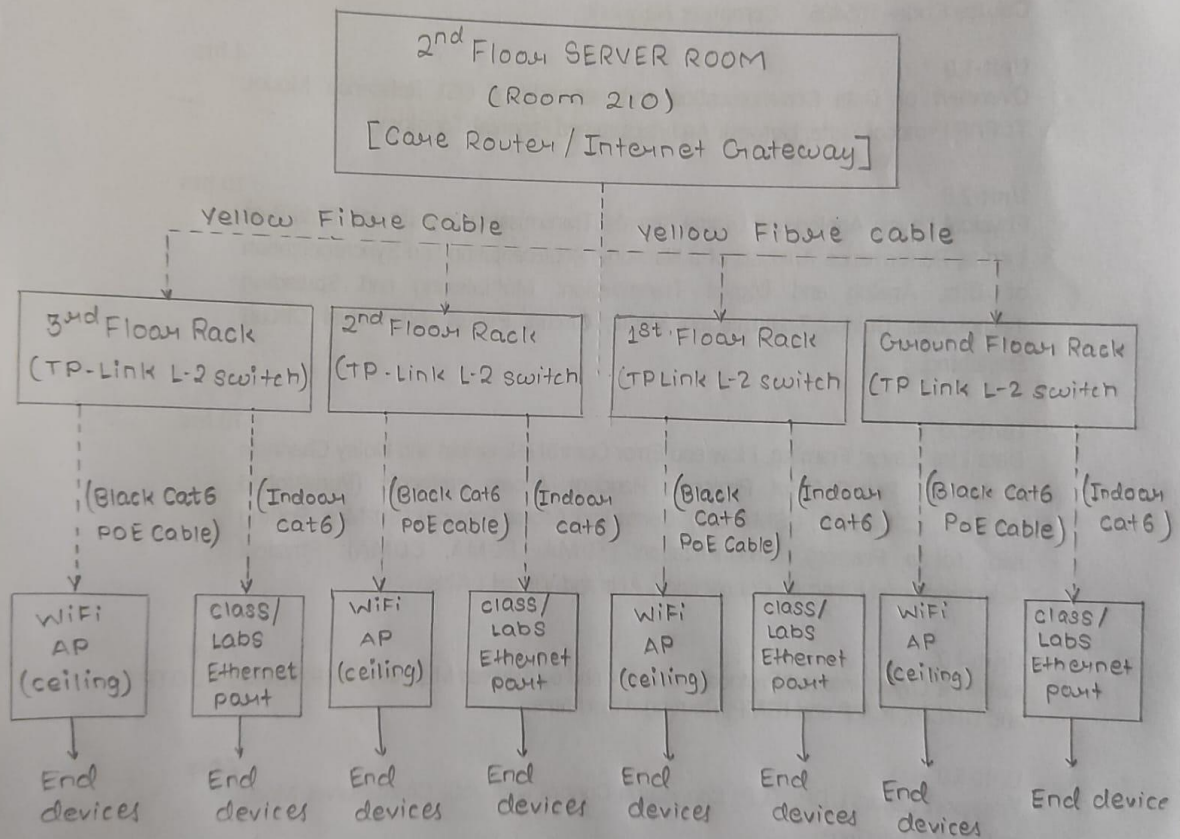
These floor switches connect back to the 2nd-floor Server Room via high-speed Fiber Optic Uplink Cables (identified by yellow patch cord connected to the SFP+ ports).

Access Layer (End Devices):

Wired Connections: Standard Cat6 Ethernet cables run through PVC conduit rising from the floor rack directly to the RJ45 wall plates/sockets inside individual classrooms and labs.

Wireless Connections: Circular Wifi Access Points (APs) are mounted on the corridor ceilings across all floors. These APs are connected to the floor's TP Link switches using Black Outdoor-Grade Cat6 Ethernet

HIERARCHICAL STAR TOPOLOGY



cables, which simultaneously carry data and electrical power via POE+ (Power over Ethernet) technology.

2. Identify Network Devices:

- List all Networking devices used in LAN.
- Mention the function of each device.

Our campus LAN uses a mix of core, routing, distribution, and access layer devices to maintain seamless connectivity. Below is the list of all networking devices identified in the LAN along with their specific functions:

i) Core Router / Main Gateway

- Device identified: High-performance Enterprise Router, 1-3 switch (located inside the 2nd Floor Server Room, Room 210).
- Function: It acts as the primary gateway between the internal college LAN and the external WAN (wide Area Network) or (Internet Service Provider).

• It handles Routing Decisions by routing data packets into and out of the campus network.

• It manages core network security, NAT (Network Address Translation), and traffic policies.

ii) L2+ Managed Switch (24-Port Gigabit with 10GE SFP+)

• Device identified: TP Link Omada SG3428XMP (Installed inside floor rack)

• Function:

• It acts as the central connection point or point for a specific floor, connecting local end-devices like PCs and WiFi Access Points to the network.

• It supports L2+ Management and VLANs (Virtual LANs) to segment network traffic (e.g., separating lab systems from administrative traffic) to prevent broadcast storms.

• It has 4-Port 10GE SFP+ slots where the yellow Fibre Optic Uplink cables are connected to deliver high-speed backbone bandwidth from the server room.

iii) Easy Smart Switch (18-Port Gigabit with 16-Port PoE+)

- Device identified: TP-Link TL-SG1218MPE (Installed inside floor racks).
- Function:
 - Used for high-density device deployment on the floors.
 - It forwards data frames based on MAC addresses to local classroom ports and labs.
 - It features PoE+ (Power over-Ethernet) capability, which supplies both electrical power and data to the WiFi access points over a single Ethernet cable.

iv) WiFi Access Point (AP)

- Device identified: Ceiling-mounted circular wireless Access Points (Distributed across corridors on all floors)
- Function:
 - It converts wired Ethernet signals into wireless radio signals (WiFi), allowing laptops, smartphones, and tablets to connect to the LAN wirelessly.

- It acts as a bridge between clients and the wired network backbone.

iv) Patch Panel

- Device identified: Multi-port metal termination panel (mounted just above the switches inside the D-Link racks).
- Functions:
 - It acts as a static cable management system. The solid Cat6 cables coming from classrooms through PVC conduits terminate at the back of the patch panel.
 - Short, flexible Patch Cords are then plugged from the front of this panel into the switch ports, ensuring neatness and preventing physical damage to the main switch ports.

vi) Network Cabinet / Rack

- Device identified: D-Link wall-mounted Networking Rack (Cabinet with a glass door).
- Function:
 - It securely encloses and protects critical hardware devices (switches).

- Patch Panels, Cables) from dust, heat, physical tampering, and accidental damage.
- Equipped with cooling fans to maintain proper operational temperature for the switches.

3. Analyze Connection Types:

- Specify whether connections are:
 - wired (Ethernet)
 - wireless (Wi-Fi)
- Mention where each type is used.

In our organization's LAN, both wired and wireless communication mediums are deployed based on bandwidth requirements, mobility and device density.

The analysis of where and why each connection type is used is detailed below:

i) Wired Connection (Ethernet)

- Medium Used: Category 6 (Cat6) Unshielded Twisted pair (UTP) cables running through horizontal PVC conduit cabling.

- Where it is used on campus : Inside Computer Labs and classrooms and Offices to connect desktop horizontal PCs and laptops to the network.
- As Uplink / Backbone connectivity to bridge different floor distribution racks back to the main Server Room on the 2nd floor.

- Why it is used here :
- High Speed & Low Latency : Wired Ethernet provides dedicated gigabit speeds (up to 1Gbps) which is crucial for heavy lab practicals, programming tasks, and stable server connections.
- No Interferences : Physical cables are immune to the radio frequency interferences caused by walls, concrete pillars, or weather changes.
- Security : It is highly secure as a user must physically plug into the RJ45 wall plate to gain access to the network.

2. Wireless Connection (Wi-Fi)

- Medium Used : IEEE 802.11 Standards (Wi-Fi)

delivered via ceiling-mounted TP-Link Omada Access Points (APs).

- Where it is used on Campus:
 - In Corridors, Verandas, and Common Areas across all three floors where continuous physical cabling to every single spot is not practical.
 - Used by students and faculty on Smartphones, Tablets, and Personal Laptops while moving around the campus.
- Why it is used here:
 - Mobility & Convenience: It allows user to stay connected to the campus network seamlessly without being physically chained to a single desk or classroom wall socket.
 - High Capacity Support: A single WiFi Access Point can handle multiple wireless clients simultaneously, avoiding the mess of hundreds of visible physical wires hanging around.
 - PoE Efficiency: As observed in the campus, these APs run on Power over Ethernet (PoE) via black

outdoor grade wires, making wireless expansion easier without laying down new power lines.

4. Measure Network Speed:

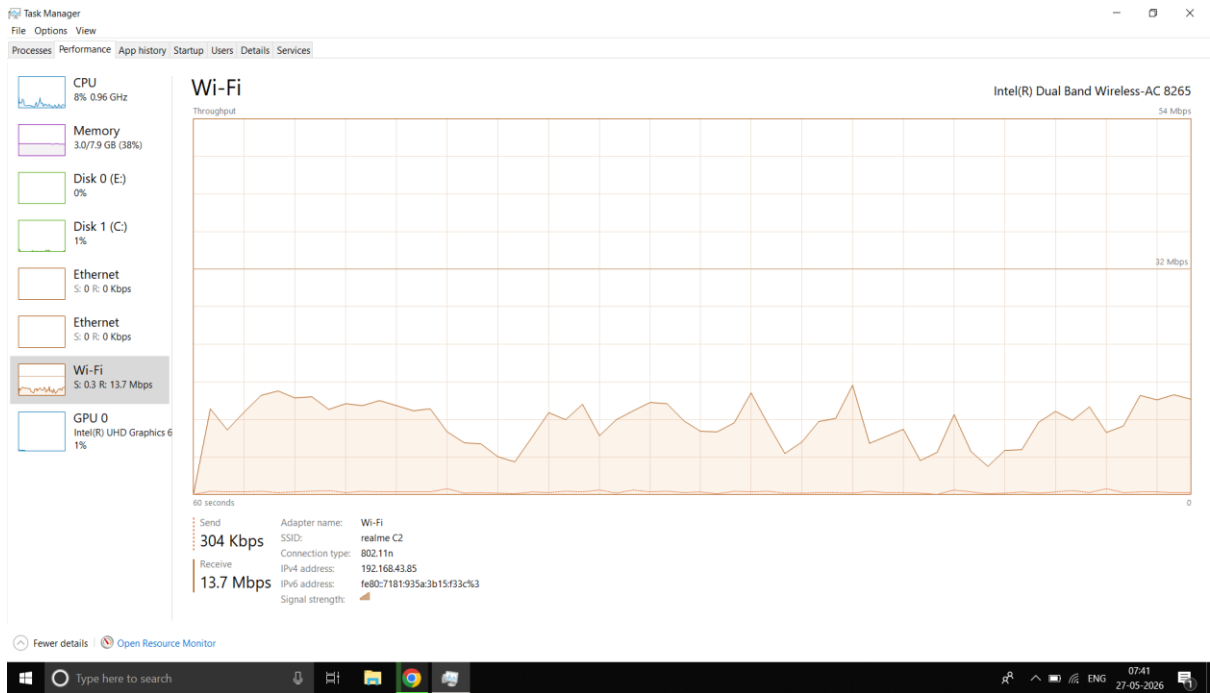
- Find the link speed of:
 - wired network (if available)
 - wireless network
- Mention the tools/commands used (e.g., Task Manager, netsh; netsh wlan show interfaces or, etc.)

5. Find System Network Details:

- Determine and record the following of your computer:
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
- Mention the command(s) used (e.g., ipconfig, getmac).

6. Explain Observations:

- Briefly explain:
 - Difference between wired and wireless speed
 - Role of subnet mask in networking.



Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

```
PS C:\WINDOWS\system32> Get-NetAdapter
```

Name	InterfaceDescription	IFIndex	Status	MacAddress	LinkSpeed
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	36	Up	00-50-56-C8-00-01	100 Mbps
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	37	Up	00-50-56-C8-00-08	100 Mbps
Wi-Fi	Intel(R) Dual Band Wireless-AC 8265	3	Up	38-BA-FB-9F-8D-AA	86.7 Mbps

```
PS C:\WINDOWS\system32> Get-NetAdapterAdvancedProperty
```

Name	DisplayName	DisplayValue	RegistryKeyWord	RegistryValue
VMware Network Adapte...	Jumbo Packet	Disabled (1514)	*JumboPacket	(1514)
VMware Network Adapte...	Priority / VLAN tag	Priority & VLAN Enabled	*PriorityVLA...	(3)
VMware Network Adapte...	Wake on magic packet	Enabled	*WakeOnMagic...	(1)
VMware Network Adapte...	Wake on pattern match	Enabled	*WakeOnPattern	(1)
VMware Network Adapte...	Wake on LAN	Enabled	*EnableWakeOnLan	(1)
VMware Network Adapte...	VLAN ID	--	VlanId	(-)
VMware Network Adapte...	Jumbo Packet	Disabled (1514)	*JumboPacket	(1514)
VMware Network Adapte...	Priority / VLAN tag	Priority & VLAN Enabled	*PriorityVLA...	(3)
VMware Network Adapte...	Wake on magic packet	Enabled	*WakeOnMagic...	(1)
VMware Network Adapte...	Wake on pattern match	Enabled	*WakeOnPattern	(1)
VMware Network Adapte...	Wake on LAN	Enabled	*EnableWakeOnLan	(1)
VMware Network Adapte...	VLAN ID	--	VlanId	(-)
Wi-Fi	Sleep on WLAN Disconnect	Disabled	*DeviceSleep...	(0)
Wi-Fi	Packet Coalescing	Enabled	*PacketCoale...	(1)
Wi-Fi	ARP offload for WLAN	Enabled	*PMASPOffload	(1)
Wi-Fi	NS offload for WLAN	Enabled	*PMNSOffload	(1)
Wi-Fi	GTK rekeying for WLAN	Enabled	*PMGSEKeyey...	(1)
Wi-Fi	Wake on Magic Packet	Enabled	*WakeOnMagic...	(1)
Wi-Fi	Wake on Pattern Match	Enabled	*WakeOnPattern	(1)
Wi-Fi	Channel Width for 2.4GHz	Auto	ChannelWidth24	(1)
Wi-Fi	Channel Width for 5GHz	Auto	ChannelWidth52	(1)
Wi-Fi	Mixed Mode Protection	RTS/CTS Enabled	CtsToItself	(0)
Wi-Fi	Fat Channel Intolerant	Disabled	FatChannelIn...	(0)
Wi-Fi	Transmit Power	5. Highest	TransmitPower	(100)
Wi-Fi	802.11n/ac Wireless Mode	802.11ac	IEEE80211Mode	(2)
Wi-Fi	MIMO Power Save Mode	Auto SPMS	MIMOPowerSav...	(0)
Wi-Fi	Roaming Aggressiveness	3. Medium	RoamingAggressi...	(2)
Wi-Fi	Preferred Band	1. No Preference	RoamingPrefer...	(0)
Wi-Fi	Throughput Booster	Disabled	ThroughputBo...	(0)
Wi-Fi	U-APSD support	Disabled	uAPSDSupport	(0)
Wi-Fi	802.11a/b/g Wireless Mode	6. Dual Band 802.11a/b/g	WirelessMode	(34)

```
PS C:\WINDOWS\system32>
```

Type here to search

ENG 08:07 27-05-2026

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> netsh wlan show interfaces

There is 1 interface on the system:

Name                : Wi-Fi
Description          : Intel(R) Dual Band Wireless-AC 8265
GUID                : 07fc3e81-d6dc-4033-8f28-28f2d5248d3c
Physical address     : 38-ba-f8-9f-8d-aa
State                : connected
SSID                : realm C2
BSSID               : 5e85e9d6f1eeff
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Auto Connect
Channel              : 6
Receive rate (Mbps) : 72.2
Transmit rate (Mbps) : 72.2
Signal               : 99%
Profile              : realm C2

Hosted network status : Not available

PS C:\WINDOWS\system32>
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . : fe80::4c18:4742:a24a:5c99%36
IPv4 Address. . . . . : 192.168.20.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . : fe80::d497:3f6d:2d1f:68e8%37
IPv4 Address. . . . . : 192.168.19.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . : fe80::7181:935a:3b15:f33c%3
IPv4 Address. . . . . : 192.168.43.85
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.43.1

PS C:\WINDOWS\system32> getmac

Physical Address    Transport Name
-----
38-BA-F8-9F-8D-AA   \Device\NPF{07FC3E81-D6DC-4033-8F28-28F2D5248D3C}
00-50-56-C0-00-01   \Device\NPF{807E1AB5-CA30-4F09-8009-966F964E8025}
00-50-56-C0-00-08   \Device\NPF{77813A2D-E0BC-481A-8998-41978A7A7AF9}

PS C:\WINDOWS\system32>
```

During the measurement phase using Windows Task Manager, a significant difference was observed between the two mediums:

- Wired Networks (Ethernet): It provided a constant link speed of 1 Gbps (1000 Mbps). Because data travels through dedicated physical pairs of copper inside the Cat6 cable, there is zero attenuation (signal drop) caused by physical walls or pillars. This makes wired connections ideal for labs where high-speed data transfer and low latency are required.
- Wireless Network (WiFi): The speed fluctuated between 144 Mbps and 16 Mbps. Wireless signals use radio frequencies travelling through the air. As a result, the speed drops significantly if a student moves away from the corridor Access Point, or if there are concrete walls/obstacles blocking the line of sight.

Role of Subnet Mask in Networking

Page No. _____
Date _____

The subnet mask observed during the test was 255.255.255.0. It plays a crucial role in the network using the following mechanisms:

- **Network vs Host Separation:** It performs a bitwise AND operation on the IP address to separate the Network ID from the Host ID. For example, in IP 192.168.10.45 with mask 255.255.255.0, the first three octets (192.168.10.x) represent the defined college network path, and the last octet (.45) identifies the specific computer.